

Graduate Diploma in Systems Analysis

Industrial Attachment Guidelines

Distribution:
All SA Students
All Participating Organisations
All NUS-ISS Advisors

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1 Introduction to the Industrial Attachment Programme

1.1 Overview

The industrial attachment programme is designed to provide students the hands-on experience to put into practice the skills, tools and techniques they have acquired on a **real world** IT project. This project provides student the opportunities to analyse, design, construct and implement a *workable* IT application system to address the business needs of a functional area in the organisation where the students are assigned.

The success of this programme will depend upon a variety of factors such as

- how well students relate to other members of the organisation so as to be an effective member of a team working towards common goals and objectives
- how effectively students manage their time so as to complete the assignments as planned
- students' abilities to:
 - communicate with others in terms of gathering facts
 - conduct interviews, and apply listening and presentation techniques
 - critically assess and identify the problem areas, state the objectives and evaluate the methodologies and techniques available in the design and construction of a workable application system

1.2 Objectives of the Industrial attachment Programme

Upon completion of the Industrial attachment programme, all students must be able to:

- describe and discuss their role and responsibilities undertaken during the industrial attachment
- relate the experience to the roles and responsibilities of a Systems Analyst
- describe the organisation and the business objectives of the function/department for which the industrial attachment project is being developed
- apply the skills acquired in the analysis, detailed design, implementation and documentation phases for a chosen application
- perform and document all deliverables in the System Development Life Cycle

1.3 Schedules for the Industrial Attachment Programme

Duration of Industrial Attachment

20 weeks	Attached to participating organisation
2 weeks after the internship period	• Prepare and complete Industrial Attachment Reports • Presentation to NUS-ISS lecturers

1.4 Important Milestones/Dates

Date	Milestones
20 weeks internship period Every Saturday	Submit Weekly plan/report to: • respective NUS-ISS Advisor's mailbox using his/her email ID • respective project folder in LumiNUS
Week 6 or 7	Email company supervisor and NUS-ISS advisor to update your progress and check if both want to schedule a meeting.
Week 17	Online briefing for Industrial attachment Presentation briefing.
Week 19	Company supervisor emails appraisal form to NUS-ISS advisor and internship manager
End of internship	Submit to Canvas
1 st week after end of the internship	Internship Report Summary for Independent Reviewer • Clearance form

1 st week after end of the internship	Submit to Canvas/Your Internship Team Final Report Folder • Industrial Attachment Report, Individual & Peer Evaluation Report • Student - Industrial Attachment Questionnaire • Poster • Presentation slides
1 st to 2 nd week after end of the internship	Industrial attachment presentation to NUS-ISS Advisor / Independent Reviewer

1.5 Leaves

After serving 3 months, the company is recommended to give students for 2 days of paid personal leaves and 3 days of paid sick leaves.

1.6 Industrial Attachment Student Work Calendar Guidelines

1.7 Week 1 to 2 (Familiarisation, Business Modelling, Requirement Capture)

- (i) Industrial attachment organisation and project orientation
- (ii) Project planning, arrange, confirm and prepare for the necessary interviews
- (iii) Review and evaluate the business functions and operations within the selected functional area; identify the key issues and interfaces
- (iv) Perform and document problem analysis
- (v) Formulate, evaluate and document alternative solutions
- (vi) Develop a conceptual outline of the proposed system
- (vii) Document requirements and conduct walkthrough with users

1.8 Weeks 3 to 5 (Analysis and Design Workflow)

- (i) Conduct detailed analysis of approved requirement
- (ii) Perform design

1.9 Week 6 to 20 (Implementation, Testing and Deployment workflow)

- (i) Perform coding and testing
- (ii) Carry out User Acceptance Testing
- (iii) Installation of new system
- (iv) Complete documentation (operation manual, user manual etc)

1.10 Weeks 21 to 22 (back to NUS-ISS)

- (i) Complete industrial attachment report at NUS-ISS
- (ii) Present Industrial attachment project to NUS-ISS advisors/lectures

Note:

Students are to use the organisation's life cycle and methodology. If the organisation does not have a specific methodology, students should use their discretion to use an appropriate methodology taught in NUS-ISS (pure OO, hybrid OO, web based methodology).

2 Responsibilities of the Industrial attachment Organisation

2.1 Organisation's Responsibilities

The organisation is responsible for providing:

- Job related project(s) which will enable the students to apply the concepts, methodologies, and techniques acquired during the course
- Supervisor(s) for each of the industrial attachment teams assigned to their organisation
- Work space and other necessary resources for the students
- Access to information and personnel required to complete the industrial attachment project

If the internship project is different from the original proposal, please inform the internship manager for approval in advance.

The organisation should refrain from engaging the students in outside-internship job during the internship period. NUS-ISS will be only responsible for the students' performance in internship project.

The organisation should allow the students to attend the compulsory events requested by the school during the internship.

The interns are protected under the MOM employment protection. The internship company should treat the interns professionally.

The internship allowance should be paid directly to the student, no less than \$800 per student/month, no CPF contribution required.

2.2 Supervisor's Responsibilities

The responsibilities of the supervisor include:

- Defining the student industrial attachment assignment
- Conducting the initial meeting with students to review the scope of the application selected and to tailor the work plan to fit the specific project
- Reviewing the progress of the students each week; to compare actual progress against planned progress and suggest corrective action, if necessary
- Supervising students and providing feedback on performance throughout the industrial attachment period
- Assisting the industrial attachment teams to arrange the required interviews
- Assisting the industrial attachment teams to gain access to organisation resources required (e.g. documentation, personnel)
- Being available to students for questions and directions. If the supervisor is unavailable, an alternative supervisor should be appointed
- Contacting the NUS-ISS advisor if questions arise regarding the student project
- Completing the student evaluation form and reviewing the evaluation with each student on an individual basis
- Ensuring proper handover and clearance of student internship report and presentation deck

3 Responsibilities of NUS-ISS Students

The role of the NUS-ISS students in the industrial attachment programme is to fulfil requirements from the Internship Company and university.

- Professionalism
 - Being courteous when interacting with the company staff and their teammates.
 - Striving themselves to produce good quality work.
 - No negotiation on their allowance
 - NUS-ISS will arrange internship companies to pay the students allowances directly. In the event, due to some unforeseen condition with valid reasons, the students should still continue working till the end of the internship.
 - No outside-internship job with the internship company or other company during the internship period.
- Self-management
 - Updating their progress to NUS-ISS advisor and company supervisor weekly
 - Working independently and manage their time wisely to meet the project deadlines
- Technical capacity
 - Delivering the project according to the company requirements which may require them to self-learn some new programming languages.
 - Having a proper hand-over before the end of internship
- Attendance
 - The NUS-ISS student should follow the company working hours.
 - Medical leaves
 - Inform the company supervisor before 10am and get a doctor MC.
 - Scan and upload the MC to your LumiNUS team folder and email you NUS-ISS advisor and cc to NUS-ISS internship manager on the 2nd day
 - When taking more than 3 days MC in total through out the internship, NUS-ISS disciplinary board will review the case and may terminate internship
 - Personal leaves
 - 2 paid of personal leaves are recommended to the internship company

- Seek approval from NUS-ISS internship manager first and then the company supervisor in advance for any urgent personal leaves
- No overseas travel allowed during internship working days except during paid-personal leaves or it has been pre-approved by NUS-ISS and the company.

4 Role of NUS-ISS Advisor

The principal role of the NUS-ISS advisor in the industrial attachment programme is to *ensure* that the students will benefit substantially from this programme.

- Each student team will be assigned an NUS-ISS staff member as their advisor for the attachment period.
- The NUS-ISS advisor will serve as a consultant to both the student teams and organisation supervisors on issues or questions related to the industrial attachment.
- The NUS-ISS advisor will review weekly project progress report submitted by students (students will email the weekly report to advisor and upload to the respective project or workbin in IVLE) and step in if necessary to ensure that the students are on the right track and that the assignment can be completed by the end of the attachment period.
- If the project turns out to be more involved and complicated than time allows, necessary action is to be taken to trim the scope of the project. This is to be done with the consent of the project supervisor.
- The NUS-ISS advisor will connect students to other NUS-ISS lecturers to seek help when necessary.
- The NUS-ISS advisor is expected to visit the site at least once and will maintain contact either via email, phone or other means with students during the industrial attachment.

5 Student and Organisation Industrial attachment Appraisals

We need both the students and the organisation to fill in the necessary questionnaires/appraisals so as to provide feedback regarding the industrial attachment assignment, these will enable NUS-ISS to improve/modify our course and industrial attachment programme. These questionnaires/appraisals are:

- Student – Industrial attachment Questionnaire (Appendix D)
- Industrial Attachment Organisation - Student Appraisal Form (Appendix E)

6 Industrial Attachment Assessment

Areas	Inputs to the assessment	Some of the Qualities Assessed
Report	Industrial attachment Report	• Completeness • Accuracy
Presentation	Presentation	• Clarity • Division of work • Conviction • Enthusiasm
Overall	Weekly Plan/Report Student Individual Report Peer Report Student – Industrial attachment Questionnaire Industrial Attachment Organisation - Student Appraisal Form NUS-ISS Advisor Evaluation	• Planning, monitoring and control of project • Teamwork • Initiative • Innovativeness • Work attitude

8 Showcase Successful Internship Projects

The school may share the successful internship projects in the NUS-ISS events and classes without revealing the sensitive data to prompt our internship programs.

Appendix A Sample Weekly Plan/Progress Report

Contents of weekly plan and progress report

Week 1 Report (DD/MM/YY- DD/MM/YY)

1. Milestones Met
2. Milestones Missed
Reason for Missing Milestones
Resolution
3. Adjustments to schedule and effort
Dates,
Background,
Results of Major Decisions
4. Foreseeable risks and resolution

Week 2 Plan (DD/MM/YY – DD/MM/YY)

1. Who will do what
2. Milestones to be met

Note

1. ***Please attach the overall project plan/schedule for the entire industrial attachment period in the first progress report***
2. ***And send the updated plan/schedule when the plan/schedule is revised subsequently.***
3. ***Send a copy of this report to your NUS-ISS Advisor***
4. ***Submit a copy of this report into your team project/workbin in Luminus***

Appendix B Sample Industrial Attachment Report

Cover page sample

SA61 INDUSTRIAL ATTACHMENT REPORT

Project Title: XX

Students: 1. XXXXXXXXX
2. XXXXXXXXX

Organisation : XX

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE
GRADUATE DIPLOMA IN SYSTEMS ANALYSIS
September 2024

NUS-ISS
NATIONAL UNIVERSITY OF SINGAPORE

Contents in the Industrial attachment Report

1. Introduction

- Project Background

2. Overview of Activities

- Project Plan and schedules
- Phases / activities / deliverables (high level description ONLY!)

3. Recommendations

This section should state any recommendations for the system and the future phases of development

- Short term recommendation
- Long term recommendation

4. Things Learned

Describe what the students have learned/gained from this industrial attachment project.

5. Problems and Solutions

This section should describe the problems encountered during the industrial attachment project and solutions to these problems, if any.

6. Looking Back

This section should describe what should have been done differently if assigned to a similar project and why?

7. Acknowledgement

8. Appendix A

Phase/Workflow	Deliverables
Business Modelling Workflow	Project Plan
	Business Use Case Model Survey
	Business Object Model Survey
Requirement Capture Workflow	Requirement Specification (Function Requirements)
	Requirement Specification (Non Functional)
	UI Design
Analysis Workflow	Analysis Models
Design Workflow	Design Models
	Relational DB Design
Implementation Workflow	Code

Phase/Workflow	Deliverables
Test Workflow	Test Case and Test Result
Deployment Workflow	Running application
Others	Other deliverables unique to the specific project/organisation

Appendix B

Progress Reports – Attach all 20 weekly progress reports, Project Plans/Schedules

Note: The report organisation (including the appendices) should always guide readers from an overview to the detailed levels. It is definitely not acceptable to expect readers to scan through the details, interlink them to achieve the overview.

The above report contents is only a guideline, students may make necessary adjustments to adhere

- ***to the organisation standard documentation if needed.***
- ***to fit the phrases and stages practiced during the attachment, as not all projects go through the entirely life cycle.***

Appendix C Student: Industrial Attachment Questionnaire

This questionnaire is to be filled out by *each* student and returned together with your project report to the NUS-ISS Advisor. All information will be held in strictest confidence. The purpose of this questionnaire is to measure the value of the industrial attachment and to identify any required modifications.

I. GENERAL

- A. Student Name _____
- B. Industrial Attachment Organisation Name: _____
- C. Industrial Attachment Organisation Supervisor: _____
- D. Industrial Attachment Dates: From _____ To _____

II. TIME ALLOWANCE

- A. Was there time allowed for the completion of your work assignments?
☐ Too Short ☐ About Right ☐ Too Long
- B. If the time was too short or too long, how much time you think should be allowed?
- C. Any other comments.

III. ASSIGNMENTS

- A. Which assignments were of the most/least value to you? Please explain.
- B. Which assignments were of the most/least value to your industrial attachment organisation? Please explain.
- C. Did the assignments make use of the knowledge/skills acquired? Please explain.

D. Which assignments, if any, would you add/delete/change? If changes how do you think they should be changed?

E. Any other comments

IV. INDUSTRIAL ATTACHMENT ORGANISATION

A. Did you get adequate guidance from your project supervisor? Please comment.

B. How much time per day did you spend on SA assignments?

C. How did you spend the working time that was not devoted to SA assignments?

D. Did you have access to all resources (people, information), needed to complete your assignments? Please explain.

E. Any other comments.

V. INDUSTRIAL ATTACHMENT ADVISOR

- A. Did your NUS-ISS advisor visit you? If YES, how many times?
- B. Were you able to contact your NUS-ISS advisor during the attachment and was the advisor prompt in responding?
- C. Did you have to contact any other NUS-ISS staff for assistance or enquiries during industrial attachment? If YES, who they were and what form of assistance did you need?
- D. Do you have any comments or compliments for your industrial attachment advisor?

VI. INDUSTRIAL ATTACHMENT OVERALL

- A. Did the industrial attachment programme meet the objectives presented in Section 1.2 of the industrial attachment guidelines?

☐

YES

☐

NO

If No, please explain.

- B. Was the industrial attachment:

☐

Very useful

☐

Some use

☐

O K

☐

Little use

☐

Useless

VII. SKILL SETS

- A. List the skill sets that is most commonly used by the staff in your industrial attachment organisation (eg. methodology, programming, networking, etc)

- B. Which of these skill sets are important to your industrial attachment project?

What other skill sets would you suggest that NUS-ISS should impart that will benefit you?

VIII.COMMENTS/SUGGESTIONS

Please make any additional comments/suggestions not specifically covered in this questionnaire. Your input on the industrial attachment programme will serve as a major source for evaluating and modifying the industrial attachment programme. Please feel free to discuss any of your comments with members of the NUS-ISS staff.

Appendix D Industrial Attachment Organisation: Student Appraisal Form

The purpose of this form is to provide feedback on student performance during the Industrial Attachment programme to the NUS-ISS staff and to the student. The supervisor who monitors the student's performance during this period should fill out the form as and when assignments are completed, and review the contents that related to student performance with the student before the end of the Industrial attachment.

You may mail this appraisal form to the NUS-ISS Advisor or place it in a sealed enveloped and give it to the student to bring it back to the advisor. We will greatly appreciate it if the appraisal form can reach NUS-ISS no later than the last day of the Industrial Attachment.

I. GENERAL

A. Industrial Attachment Organisation Name: _____

B. Responsible Supervisor Name: _____

C. Supervisor Designation: _____

Business Phone #: _____

Email: _____

D. Student Names:

(1) _____ (2) _____

E. Industrial Attachment Dates:

From _____ To _____

II. ASSIGNMENTS

A. APPLICATION

Was the application the same as the one proposed for Industrial Attachment? If not, state the nature of the new application.

B. PROJECT SCOPE

1. How large was the assignment in terms of number of:

Program Modules _____

Batch Reports _____

Screens _____

Others _____

2. How much effort (time) was required to reduce the scope of the project to match the 19 week time periodif at all?

3. Did the students have enough time to complete the assignment in the time allowed? Please comment.

4. What is the programming language & methodology used in the project?

C. APPLICATION ANALYSIS

1. Please comment on the thoroughness and accuracy of the Analysis portion of the assignment.

2. Completion date for Application Analysis:_____

D. SYSTEMS DESIGN

1. Please comment on the quality of the Systems Design completed by the students in terms of:
 - (i) Technical feasibility/accuracy...will the system work?

(ii) Has the system been installed/implemented?

(iii) Is the system easy to use (by User Department)?

(iv) Is the documentation readable and maintainable?

E. OVERALL EVALUATION

1. Which assignments were most useful to your organisation?

2. What is your overall assessment of the students' work?

☐ Very Good

☐ Good

☐ Average

☐ Poor

Additional comments: _____

3. Based on the industrial attachment's objectives, as you understand them, do you feel the students met those objectives?

☐ Yes

☐ Some-what

☐ No

4. Please evaluate quantitatively the contributions from the interns. Each project team is given a total of 100 points that needs to be allocated amongst the team members based on the contribution of each member.

Team Member's Name	Points
TOTAL	100

F. STUDENT INDIVIDUAL EVALUATION

Please rate each student performance in the areas of taking initiative, technical capacity, self-management, quality of work and added value.

Student 1: _____

Total Leaves taken: MC _____ days Personal Leave _____ days

Initiative:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Resourcefulness:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Team Player:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Willingness of Picking Up New Technologies/Assignments:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Analysis & Design:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Programming:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Project / Tasks Management:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Self-Management / Discipline:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Quality of Work:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Added Value:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Student 2: _____

Total Leaves taken: MC _____ days Personal Leave _____ days

Initiative:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Resourcefulness:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Team Player:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Willingness of Picking Up New Technologies/Assignments:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Analysis & Design:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Programming:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Project / Tasks Management:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Self-Management / Discipline:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Quality of Work:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
☐ Acceptable, but can be improved

Added Value:

- ☐ Excellent ☐ Very Good ☐ Good ☐ Fair
- ☐ Acceptable, but can be improved

III. STUDENT WORK HABITS

- A. What was the level of professionalism* displayed by the students throughout the industrial attachment?
* Professionalism includes judgement, relationship with others, and constructive use of time.

- B. How well did the students function as a team? Did they complement each other's work effort (i.e., supportive of each other)? Please comment.

- C. Did one member of the student team stand out as the team leader? Which student? Please comment.

- D. Was the workload evenly divided between the students? Please comment.

- E. How much assistance/guidance did the students require from you?

- F. Did the students display a high/medium/low level of initiative in completing assignments?

- G. How well did the students get along with peers/supervisors during the period? Did the students generate respect from others while carrying out assignments? Please comment.

IV. SKILL SETS

What skill sets would you suggest NUS-ISS to include in our curriculum to best prepare our students for the industry (e.g. methodology, programming, networking, etc)?

V. COMMENTS/SUGGESTIONS

Please make any comments or suggestions that you feel are pertinent to the industrial attachment period (e.g. *guidelines, assignments, assessments, etc*).

(We depend upon your views as a major source of input in shaping our future industrial attachment programme)

Supervisor Name _____

Signature/date _____

[Note. The interns will forward a softcopy of Appendix E two weeks before the end of the internship, for the project supervisor to fill in.]

Appendix E Industrial Attachment Presentation Guidelines

Students should use these guidelines in organising and preparing for their industrial attachment presentation.

- A. **Presentation Type.** PowerPoint presentation of Industrial attachment results. Audience includes NUS-ISS staff members and fellow students. A demonstration of the product developed to be included. Students should seek permission from the industrial attachment organisation for the demonstration.
- B. **Schedule.** 30 minutes is allocated for each team with 2 members. The presentation should be prepared to last 20 to 25 minutes, including time for questions.
- C. **Content.** Your presentation is to describe the benefits derived by yourselves and by the industrial attachment organisation as a result of the Industrial attachment Programme. Points that might be included are:
 - Project Background
 - Business value of the project
 - Major Features developed or delivered
 - Results/products of Industrial attachment
 - Benefits derived by organisation including value added from the project
 - A 3 min demo of the application build (if any)
 - Live demo if possible, else screen grabs

You can organise your presentation in any manner you feel is most effective. Keep in mind that your major purpose is to educate your audience on the benefits of the Industrial attachment Programme. Some organisation background may be necessary for them to fully appreciate your presentation.

Note: Your Industrial attachment report will contain details that are not required to be presented, i.e. select your presentation material from the report with caution.

- D. **Participation.** Each student should participate in the organisation, preparation and delivery of the presentation on an equal basis.
- E. **Students are responsible for ensuring that only legal software presentation packages are used.** Students must install and configure any software required for their presentation.

Students will be held equally responsible for the organisation and content of the presentation. Personal style will be evaluated on an individual basis for the part delivered.

Each student is responsible for understanding the entire report and will be expected to actively participate in answering questions at the conclusion of the presentation.